

MiniRAG **MSA** Contract Q&A System

Retrieval-Augmented Generation for India Post Payments Bank

MSA

AI Interop Engineer Project

SYSTEM OVERVIEW

A full-stack RAG application built to extract, analyze, and evaluate Master Service Agreements with high precision and legal grounding.

CORE REQUIREMENTS



Ingestion

Automatic PDF extraction and recursive text chunking for legal clauses.



Retrieval

FAISS-based vector search with query expansion for contract terms.



Generation

Local LLM (Qwen) grounded answers strictly from retrieved MSA context.



Evaluation

Automated judging against 10 ground-truth legal categories.

Overall Workflow

Streamlined Document Analysis and Retrieval Process



THE TECHNICAL STACK

Frontend & Orchestration

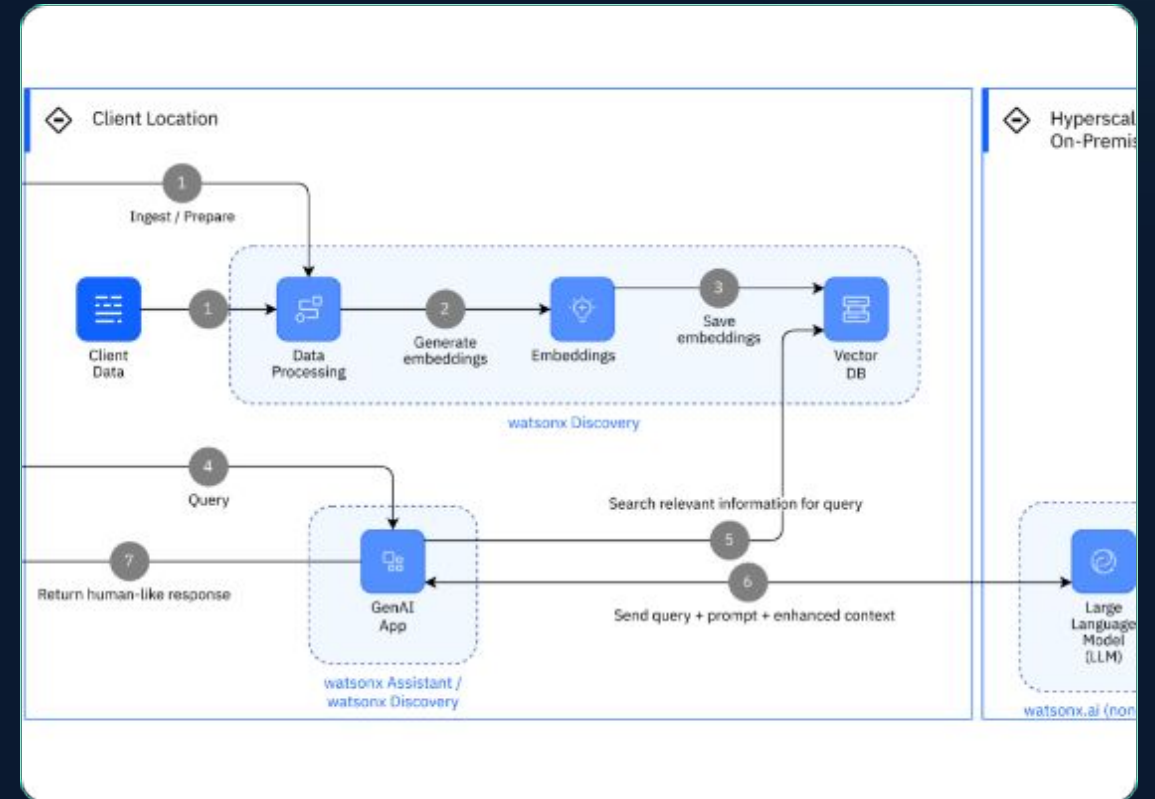
- ✓ **Django**: Robust web UI and API routing.
- ✓ **pypdf**: Seamless extraction of 24-page PDF data.
- ✓ **Langchain**: Intelligent chunking and Embedding.

Vector Intelligence & LLM

- ✓ **FAISS**: High-performance similarity indexing.
- ✓ **SentenceTransformers**: mpnet-base-v2 embeddings.
- ✓ **Ollama / Qwen**: Privacy-first local LLM inference.

INGESTION PIPELINE

- **Extraction:** Converts PDF to plain text preserving page metadata.
- **Vectorization:** Each chunk is converted into a 768-dim dense vector.
- **FAISS Sync:** In-memory vector store refreshes automatically after every upload.
- **Persistence:** Indexes and source documents are stored for session reliability.



OPTIMIZED CHUNKING

1200

Character Chunk Size

200

Character Overlap

Preserving Legal Context

Legal clauses are often dense and spans multiple sentences. A 1200-char size ensures notice periods, caps, and remedies stay together. The 200-char overlap prevents information loss at clause boundaries.

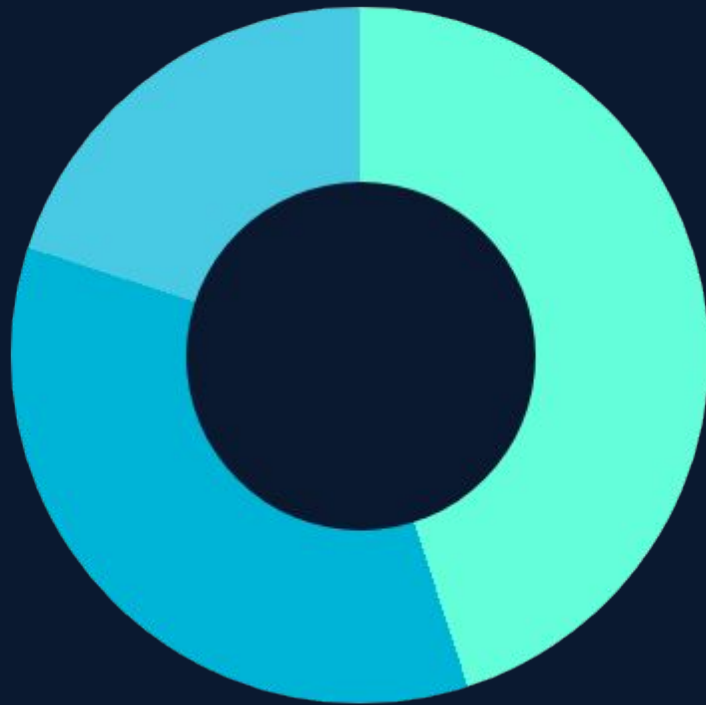
ADVANCED RETRIEVAL



Query Expansion & Reranking

-  **Synonym Mapping:** Expands "Who owns IP?" to include "Proprietary Rights" and "Deliverables".
-  **Keyword Reranking:** Boosts chunks containing exact legal terms like "Arbitration" or "Jurisdiction".
-  **Signal Detection:** Classifies similarity as Strong (0.85+), Good, Weak, or Poor.

RETRIEVAL CONFIDENCE



- Strong (0.85+): Direct clause matches.
- Good (0.70-0.84): Relevant semantic context.
- Weak / Poor: Noisy or unrelated content.

The system displays a **Similarity Report** with ranked evidence for every answer.

ScreenShots

[MiniRAG Contract QA](#)

● MiniRAG Contract QA

Ask sharper questions about dense MSA contracts.

Upload a contract, retrieve the most relevant clauses, and generate grounded answers with source similarity signals for review.

Start with Upload ->

Open QA Workspace

View Evaluation

01

Upload

Load the MSA document that powers retrieval for the question answering flow.

02

Ask

Submit contract questions and inspect the retrieved chunks behind each answer.

03

Evaluate

Compare generated answers against ground truth to measure system reliability.

Answer Trace

Grounded

QUESTION

What are the termination rights under the master service agreement?

1

Page 8

Termination for material breach

0.91

2

Page 11

Notice period and cure window

0.84

3

Page 14

Survival of obligations

0.78

GENERATED ANSWER

The agreement allows termination after written notice if a material breach is not cured within the stated notice period.

ScreenShots

[MiniRAG Contract QA](#)

Upload MSA Contract

Upload MSA: Choose File  ip_msa.pdf

Upload & Process

ScreenShots

MiniRAG Contract QA

Ask Contract Questions

What is the penalty for the provider missing a delivery milestone



Ask Question

Similarity Report

Rank	Location	Similarity	Signal	Preview
1	Page 15	0.50	Weak	34. LIQUIDATED DAMAGES If the Service Provider fails to comply with the terms of this agreement, IPPB shall, witho...
2	Page 4	0.48	Poor	In the event of the Service Provider being unable to fulfil its obligation under the Agreement or cause delay in pro...
3	Page 15	0.48	Poor	Provider, as the case may be will have the option to terminate the Agreement. Any notice served pursuant to this Cla...
4	Page 20	0.48	Poor	per the mutually agreed terms & conditions. In addition to the cancellation of the purchase order, the IPPB also res...
5	Page 16	0.48	Poor	contract, the vendor would not be liable for imposition of any such sanction so long as the delay and/or failure of...

Generated Answer

The penalty for missing a delivery milestone (delay) is a deduction from the order value of 0.5% of the services for each week or part thereof of delay, up to a maximum deduction of 10% of the order value. Once the maximum is reached, IPPB may consider cancellation of the order or termination of the agreement, and the Performance Security submitted may be invoked and forfeited.

Run Evaluation

ScreenShots

[MINIRAG Contract QA](#)

Ground Truth Evaluation

Run Evaluation

Evaluation Summary

Category	Judgement	Reason
Payment Terms	No Match	: The system answer is empty and contradicts the ground truth, as it states the MSA didn't specify a fixed invoice due date but fails to provide any details.
Late Payment Penalty	No Match	: The system answer is empty, contradicts the ground truth, or gives the wrong legal fact. The ground truth states the MSA does not specify penalties/rates, while the system answer mentions inability to find the answer, which doesn't align with the legal facts.
Delivery Deadline Penalty	Match	: The system answer contains the same key legal facts, even if wording is different. The deduction amounts (0.5% and 10%) and the cap are correctly mentioned, matching the ground truth's details.
Termination Conditions	Match	: The system answer contains the same key legal facts as the ground truth answer, which includes conditions like non-delivery, service level breaches, financial losses, and service provider bankruptcy. Both answers mention the service provider's inability to terminate for convenience, which is explicitly stated in the ground truth.
Termination Notice Period	Match	: The system answer contains the same key legal facts, even if wording is different. The ground truth mentions that the exact convenience termination notice period isn't filled, and the system answer states that the answer isn't found in the contract text.
Limitation of Liability	Match	: The system answer contains the key legal fact that aggregate liability is limited to the total Contract Price, but it misses important qualifiers such as exclusions for indemnity, breach of IP, or confidentiality obligations.
Intellectual Property	Match	: The system answer contains the same key legal facts as the ground truth answer. Both mention that all deliverables and intellectual property rights are proprietary to IPPB, and that IPPB retains ownership of IP created under the agreement.
Confidentiality Duration	No Match	: The system answer is empty, contradicts the ground truth, or gives the wrong legal fact.
Governing Law	Match	: The system answer mentions India, which is the correct country, but fails to include the specific legal facts required (governing laws and exclusive jurisdiction at Delhi).
Dispute Resolution	Match	: The system answer mentions the Indian Arbitration and Conciliation Act (1996) as part of the process, which matches the ground truth's reference to this legal fact. However, it misses the key qualifier of "escalate to senior authorized personnel" and the step of appointing arbitrators through mutual consent, which are essential components of the process outlined in the ground truth.

Accuracy: 70%

FUTURE ENHANCEMENTS



Hybrid Search

Combine FAISS with BM25 for keyword precision.



Clause-Aware

Splitting text by numbered contract headings.



Cross-Encoder

Deep neural reranking of top-20 candidates.



Page Citations

In-line citations with direct links to PDF pages.

QUESTIONS & DISCUSSION



Thank you for your time. The MiniRAG system is ready for testing.